1. **Circular polarized wave problem:**

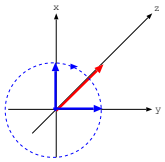
a)

i. (1.5 pts.) The electric field phasor $\tilde{\mathbf{E}} = (2\hat{x} + j2\hat{y})\, e^{-j\beta z}$ can also be written as $\tilde{\mathbf{E}}\,|_{z=0} = 2\hat{x} + 2\hat{y}e^{j\frac{\pi}{2}}$. This simply shows that $\hat{x}$ component lags s the $\hat{y}$ component by 90°.

ii. (1.5 pts.) At $z = 0$, the wave field vector is defined by

$$\mathbf{E} = \mathrm{Re}\{(2\hat{x} + j2\hat{y})e^{j\omega t}\} = \hat{x}2\cos(\omega t) - \hat{y}2\sin(\omega t).$$

The wave propagates in the $+z$-direction (as the phase $e^{-j\beta z}$ indicates). Also, $\hat{x}$ component lags the $\hat{y}$ component. Hence, the vector $\mathbf{E}$ rotates from x- toward y-axis. Therefore, the wave is **left-hand circularly** polarized.



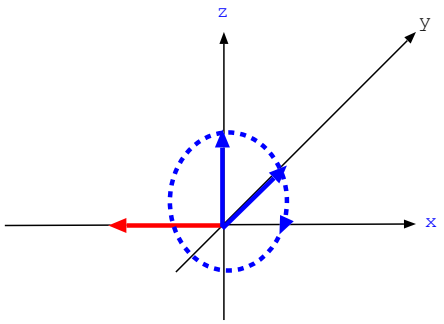
b) (3 pts.) The magnetic intensity phasor $\tilde{\mathbf{H}} = (\hat{z} + j\hat{y})\, e^{j\beta x}$ can also be written as

$$\tilde{\mathbf{H}}\,|_{x=0} = \hat{z} + \hat{y}e^{j\frac{\pi}{2}},$$

whereas the magnetic field vector is defined by

$$\mathbf{H}\,|_{x=0} = \mathrm{Re}\{(\hat{z} + j\hat{y})e^{j\omega t}\} = \hat{z}\cos(\omega t) - \hat{y}\sin(\omega t).$$

The propagation of the wave is along $-x$-direction. You can easily see that $\hat{y}$ is the leading component of the wave. As suggested in the hint, in circularly polarized waves $\mathbf{E}$ and $\mathbf{H}$ vectors will always rotate in the same direction. Therefore, the vector $\mathbf{H}$ rotates from z- toward -y-axis. Then, we can say that the $\tilde{\mathbf{H}}$ describes a **LHC** TEM wave.



c) (4 pts.)

i. $\mathbf{E}_1 = 4\sin(\omega t - \beta z)\,\hat{x}$ V/m is linear polarized in $\hat{x}$direction.

ii. $\mathbf{E}_2 = 5\cos(\omega t - \beta z)\,\hat{x} - 5\cos(\omega t - \beta z)\,\hat{y}$ V/m is linear polarized in $(\hat{x} - \hat{y})$ direction.

iii. $\mathbf{H}_3 = 2\cos(\omega t + \beta z + \frac{\pi}{3})\,\hat{x} - 2\cos(\omega t + \beta z - \frac{\pi}{6})\,\hat{y}$ A/m is circular polarized, RHC.

iv. $\mathbf{H}_4 = 4\cos(\omega t + \beta x)\,\hat{z} + 4\sin(\omega t + \beta x)\,\hat{y}$ A/m is circular polarized, RHC.

2. **(Bonus Problem: 10pts) Wave polarizations:**
The phasors form of surface current densities are(2 pts)

$$\tilde{\mathbf{J}}_{s1} = \hat{z} J_1 e^{j\phi}\,\frac{\text{A}}{\text{m}} \quad (x = 0),$$

$$\tilde{\mathbf{J}}_{s2} = \hat{y} J_2\,\frac{\text{A}}{\text{m}} \quad (x = \frac{\lambda}{4}).$$

Then, recalling $\beta = \frac{2\pi}{\lambda}$, the corresponding electric fields propagating in the region $x > \frac{\lambda}{4}$ are given by(2 pts)

$$\begin{aligned}
\tilde{\mathbf{E}}_1 &= -\hat{z}\frac{\eta_o}{2} J_1 e^{j\phi} e^{-j\beta x} = \hat{z}\frac{\eta_o}{2} J_1 e^{j(-\beta x + \phi + \pi)}\,\frac{\text{V}}{\text{m}}, \\
\tilde{\mathbf{E}}_2 &= -\hat{y}\frac{\eta_o}{2} J_2 e^{-j\beta(x - \frac{\lambda}{4})} = \hat{y}\frac{\eta_o}{2} J_2 e^{j(-\beta x + \frac{3\pi}{2})}\,\frac{\text{V}}{\text{m}}.
\end{aligned}$$

The total field is $\tilde{\mathbf{E}} = \tilde{\mathbf{E}}_1 + \tilde{\mathbf{E}}_2 = \frac{\eta_o}{2} J e^{j(-\beta x + \pi)} \left[\hat{y} e^{j\frac{\pi}{2}} + \hat{z} e^{j\phi}\right]$ where $J = J_1 = J_2$.

a) (2 pts)The direction of propagation is $+\hat{x}$ and the wave is **RHC**. Then, the $\hat{y}$ component needs to lead the $\hat{z}$ component by 90°. Therefore

$$\phi = \frac{\pi}{2} - \frac{\pi}{2} + 2n\pi = 2n\pi,$$

where n is an arbitrary integer. The electric field phasor for the region will be

$$\tilde{\mathbf{E}} = \frac{\eta_o}{2} J e^{j(-\beta x + \pi)} \left[\hat{y} e^{j\frac{\pi}{2}} + \hat{z} e^{j2n\pi}\right] \frac{\text{V}}{\text{m}}.$$

b) (2 pts)To have left handed circular polarization, the $\hat{z}$ component needs to lead by 90° the $\hat{y}$ component. Therefore

$$\phi = \frac{\pi}{2} + \frac{\pi}{2} + 2n\pi = \pi + 2n\pi,$$

where n is an arbitrary integer. The electric field phasor for the region will be

$$\tilde{\mathbf{E}} = \frac{\eta_o}{2} J e^{j(-\beta x + \pi)} \left[\hat{y} e^{j\frac{\pi}{2}} - \hat{z} e^{j2n\pi}\right] \frac{\text{V}}{\text{m}}.$$

c) (2 pts)To have linear polarization, the $\hat{z}$ component needs to be in phase with the $\hat{y}$ component or off by 180°. Thus, we write

$$\phi = \frac{\pi}{2} + n\pi,$$

where n is an arbitrary integer. The electric field phasor for the region will be

$$\tilde{\mathbf{E}} = \frac{\eta_o}{2} J e^{j(-\beta x + \pi)} \left[\hat{y} e^{j\frac{\pi}{2}} + \hat{z} j e^{jn\pi}\right] \frac{\text{V}}{\text{m}}.$$

3. **Plane wave reflections :**
The phasor form of the incident wave is $\tilde{\mathbf{E}}_i = -j4e^{-j\beta_1 z}\hat{y} = 4e^{-j\beta_1 z} e^{-j\frac{\pi}{2}}\hat{y}\,\frac{\text{V}}{\text{m}}$.

a) (3 pts.) The reflection coefficient is given by

$$\Gamma = \frac{\eta_2 - \eta_1}{\eta_2 + \eta_1} = -\frac{1}{5} \quad \text{and} \quad \tau = 1 + \Gamma = \frac{4}{5}.$$

Then, the reflected and transmitted electric fields are expressed as

$$\tilde{\mathbf{E}}_r = \Gamma 4 e^{j\beta_1 z} e^{-j\frac{\pi}{2}} \hat{y} = -\frac{4}{5} e^{j\beta_1 z} e^{-j\frac{\pi}{2}} \hat{y} \, \frac{\mathrm{V}}{\mathrm{m}}$$

and

$$\tilde{\mathbf{E}}_t = \tau 4 e^{-j\beta_2 z} e^{-j\frac{\pi}{2}} \hat{y} = \frac{16}{5} e^{-j\beta_2 z} e^{-j\frac{\pi}{2}} \hat{y} \, \frac{\mathrm{V}}{\mathrm{m}}$$

where $\beta_1 = \omega\sqrt{\epsilon_o \mu_o} = \frac{\omega}{c}$ and $\beta_2 = \omega\sqrt{\frac{9}{4}\epsilon_o \mu_o} = \frac{3}{2}\frac{\omega}{c}$.

b) (3 pts.) As the incident wave is TEM, we can solve for the associated magnetic fields by referring to $\tilde{\mathbf{H}} = \frac{\hat{n} \times \tilde{\mathbf{E}}}{\eta}$ which is given in a general form. The incident, reflected, and transmitted magnetic fields are given by

$$\begin{aligned} \tilde{\mathbf{H}}_i &= -\frac{1}{30\pi} e^{-j\beta_1 z} e^{-j\frac{\pi}{2}} \hat{x} \, \frac{\mathrm{A}}{\mathrm{m}}, \\ \tilde{\mathbf{H}}_r &= -\frac{1}{150\pi} e^{j\beta_1 z} e^{-j\frac{\pi}{2}} \hat{x} \, \frac{\mathrm{A}}{\mathrm{m}}, \end{aligned}$$

and

$$\tilde{\mathbf{H}}_t = -\frac{1}{25\pi} e^{-j\beta_2 z} e^{-j\frac{\pi}{2}} \hat{x} \, \frac{\mathrm{A}}{\mathrm{m}},$$

respectively.

c) (4 pts.) The time-average Poynting vectors of the incident, reflected, and transmitted waves are given by

$$\begin{aligned} \langle \mathbf{S}_i \rangle &= \frac{1}{2} \mathrm{Re}\left\{ \tilde{\mathbf{E}}_i \times \tilde{\mathbf{H}}_i{}^* \right\} = \frac{8}{\eta_o} \hat{z} \, \frac{\mathrm{W}}{\mathrm{m}^2}, \\ \langle \mathbf{S}_r \rangle &= \frac{1}{2} \mathrm{Re}\left\{ \tilde{\mathbf{E}}_r \times \tilde{\mathbf{H}}_r{}^* \right\} = -\frac{8}{25\eta_o} \hat{z} \, \frac{\mathrm{W}}{\mathrm{m}^2}, \\ \langle \mathbf{S}_t \rangle &= \frac{1}{2} \mathrm{Re}\left\{ \tilde{\mathbf{E}}_t \times \tilde{\mathbf{H}}_t{}^* \right\} = \frac{16}{\eta_2} \tau^2 \hat{z} = \frac{192}{25\eta_o} \hat{z} \, \frac{\mathrm{W}}{\mathrm{m}^2}, \end{aligned}$$

respectively. As it can be seen, the power density calculated for the incident, reflected, and transported waves will satisfy

$$|\langle \mathbf{S}_i \rangle| = |\langle \mathbf{S}_r \rangle| + |\langle \mathbf{S}_t \rangle|,$$

which shows that the calculations are in compliance with energy conservation principle. Also, note that

$$\Gamma^2 + \frac{\eta_1}{\eta_2} \tau^2 = 1.$$

4. (10 pts.) **Telegrapher's equations :**
The voltage and current waves $V(z,t)$ and $I(z,t)$ that propagate on a transmission line satisfy the following set of partial differential equations (PDE's)

$$-\frac{\partial V}{\partial z} = \mathcal{L} \frac{\partial I}{\partial t}, \tag{1}$$

$$-\frac{\partial I}{\partial z} = \mathcal{C} \frac{\partial V}{\partial t}. \tag{2}$$

Given that $V(z,t) = 6\cos(\omega t + \beta z)$, we get

$$\frac{\partial I}{\partial t} = -\frac{1}{\mathcal{L}}\frac{\partial V}{\partial z} = 6\frac{\beta}{\mathcal{L}}\sin(\omega t + \beta z),$$

by utilizing (1). Then, after integrating it over time, we get

$$I = -6\frac{\beta}{\omega\mathcal{L}}\cos(\omega t + \beta z).$$

Inserting this result into (2), we get

$$\frac{\partial V}{\partial t} = -\frac{1}{\mathcal{C}}\frac{\partial I}{\partial z} = -6\frac{\beta^2}{\omega\mathcal{L}\mathcal{C}}\sin(\omega t + \beta z),$$

and integrating over time, we finally obtain

$$V = 6\,\frac{\beta^2}{\omega^2\mathcal{L}\mathcal{C}}\cos(\omega t + \beta z).$$

This result implies that

$$\frac{\beta^2}{\omega^2\mathcal{L}\mathcal{C}} = 1, \qquad \text{then} \qquad \beta = \omega\sqrt{\mathcal{L}\mathcal{C}}.$$

In addition, the expression for the current becomes

$$I = -6\sqrt{\frac{\mathcal{C}}{\mathcal{L}}}\cos(\omega t + \beta z).$$